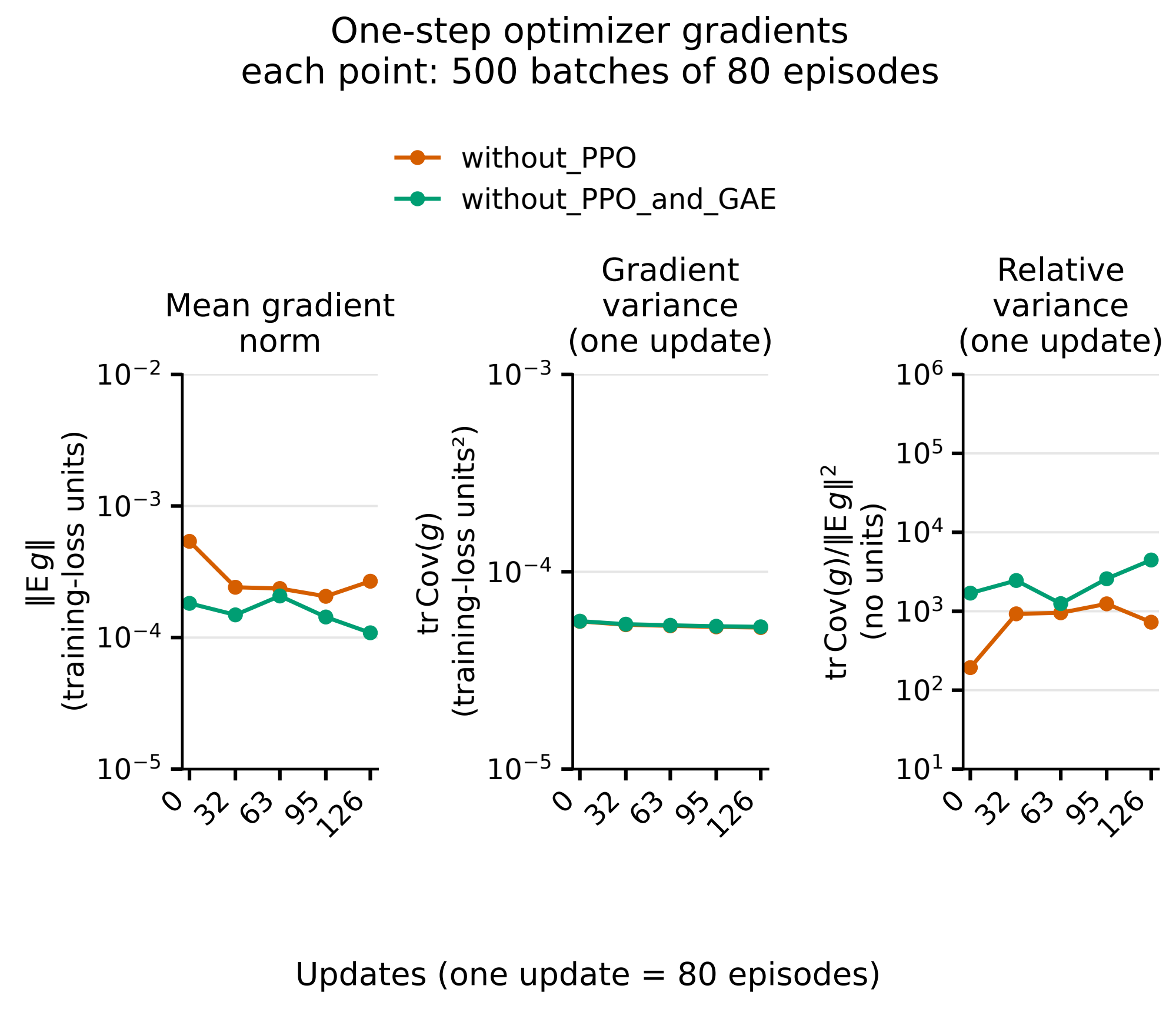
One-step optimizer gradients
each point: 500 batches of 80 episodes
without_PPO
without_PPO_and_GAE
Mean gradient norm
Gradient variance (one update)
Relative variance (one update)
‖E g‖ (training-loss units)
tr Cov(g) (training-loss units²)
tr Cov(g)/‖E g‖² (no units)
10⁻²
10⁻³
10⁻⁴
10⁻⁵
10⁻³
10⁻⁴
10⁻⁵
10⁶
10⁵
10⁴
10³
10²
10¹
0
32
63
95
126
Updates (one update = 80 episodes)